

# Application Flow Document

## Hospital Bed Allocation System

**Version:** 1.0

**Date:** 2026-05-17

**Purpose:** Define complete user journeys, system flows, and state transitions

### 1. User Personas

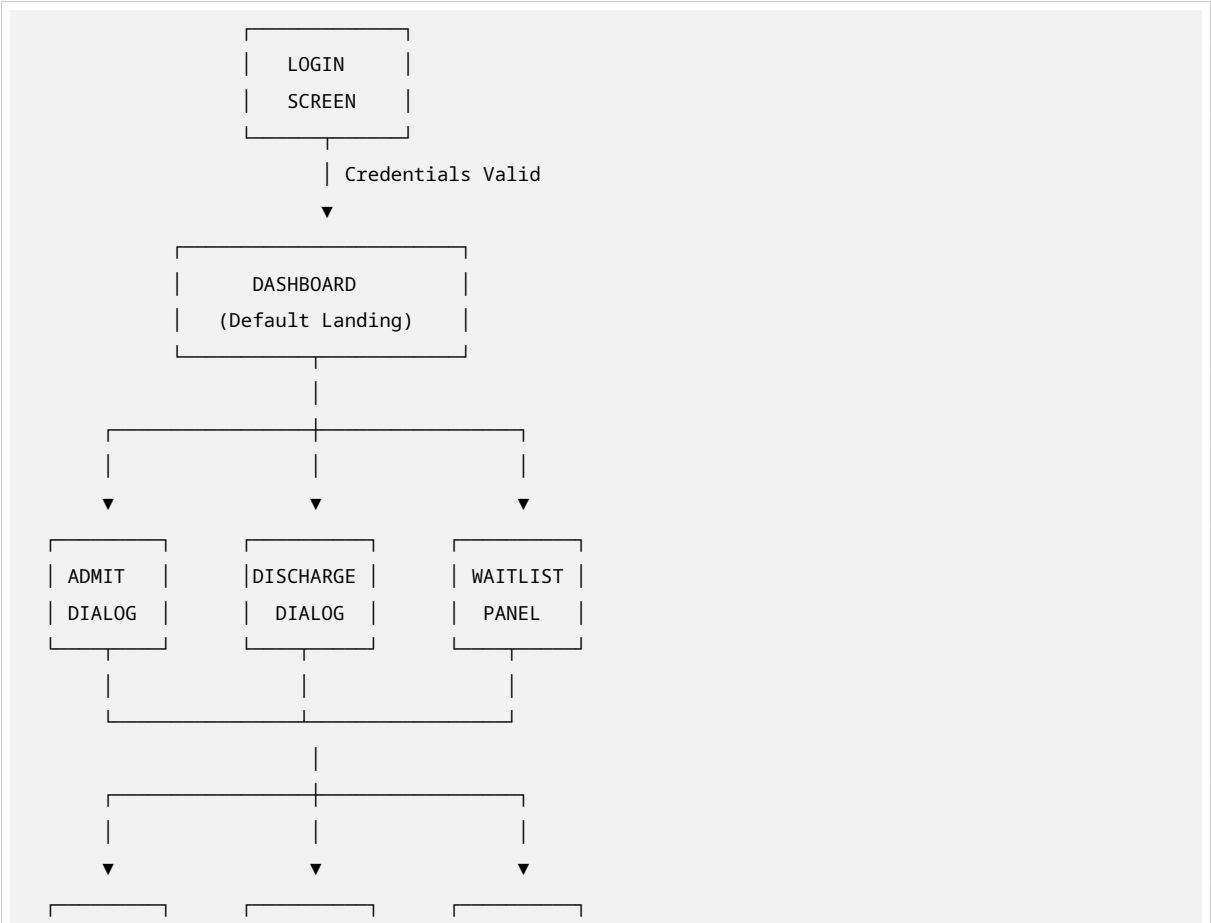
#### 1.1 Receptionist (Primary User)

- Role:** Admit patients, process discharges, view bed status, manage waitlist
- Access:** Dashboard, Admit, Discharge, Waitlist, Patients, Reports (read-only)
- Goal:** Fast, error-free bed allocation without double-booking

#### 1.2 Administrator

- Role:** Configure wards/beds, manage staff, view all reports, handle maintenance
- Access:** All screens including Admin Panel
- Goal:** Maintain accurate inventory and generate analytics

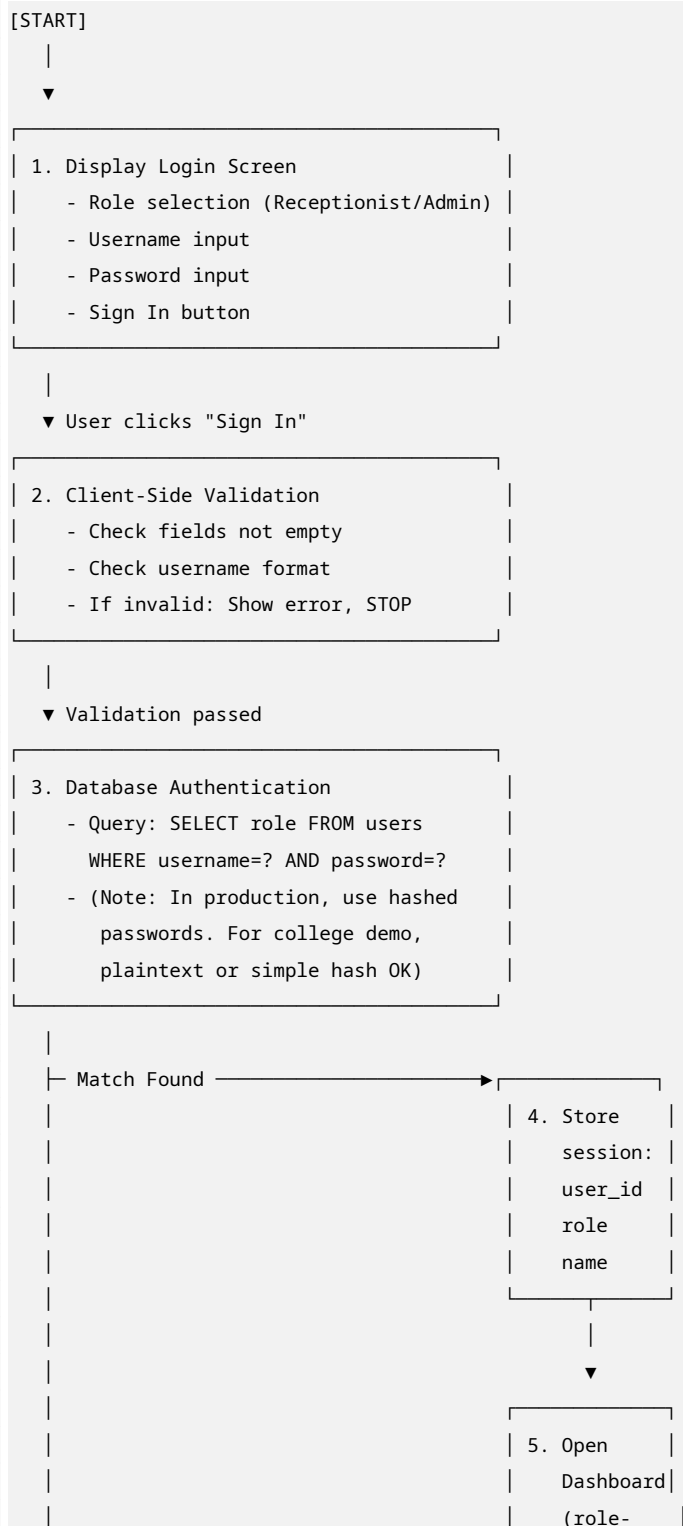
### 2. Global Navigation Flow



|          |         |       |
|----------|---------|-------|
| PATIENTS | REPORTS | ADMIN |
| MANAGER  | CENTER  | PANEL |

### 3. Detailed Flows

#### 3.1 Flow 1: User Authentication





- Doctor: "Dr. Rajesh Sharma"
- Severity: 4 (High)
- Admission Date: Today
- Expected Discharge: 2026-05-20
- Notes: "Chest pain, needs monitor"

|

▼ User clicks "Admit Patient"

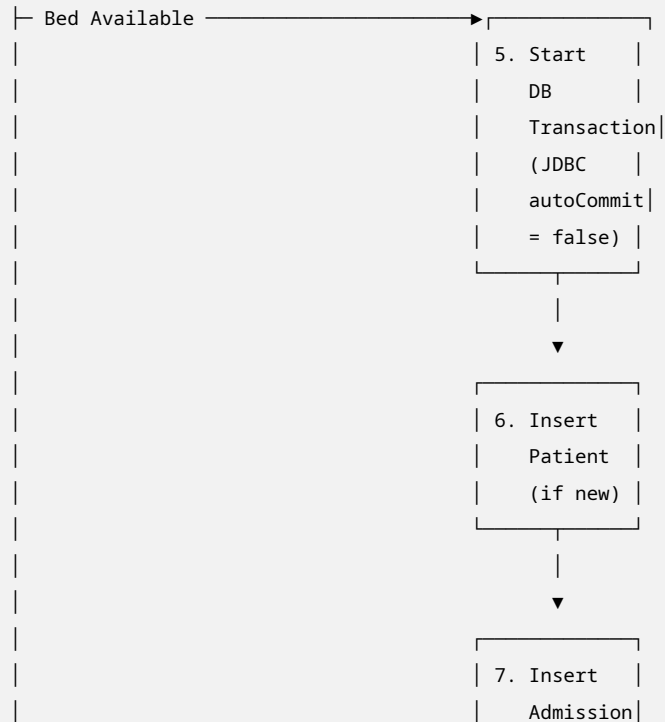
3. Real-Time Validation
- Name: Not empty, min 2 chars
  - Age: 0-150
  - Contact: 10 digits
  - Ward Type: Selected
  - Doctor: Selected
  - Expected > Admission date
  - If invalid: Red border + message
  - Scroll to first error

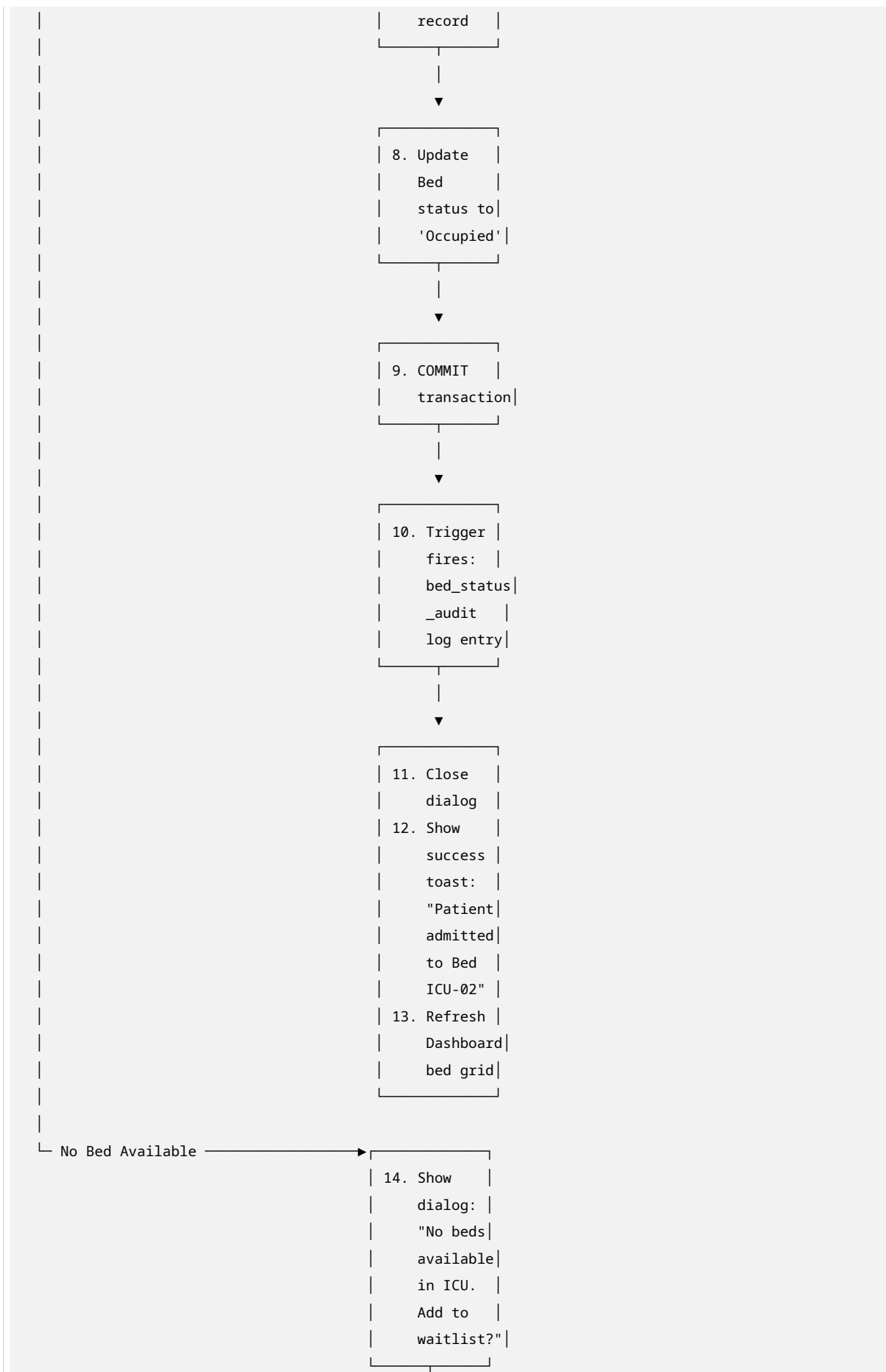
|

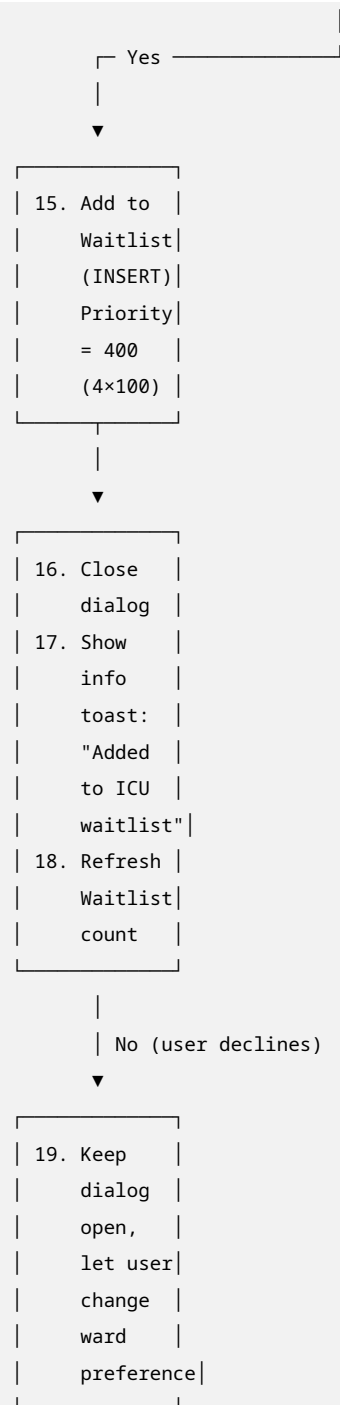
▼ Validation passed

4. Check Bed Availability
- If preferred bed specified:  
Check if that specific bed is Available
  - Else: Find first Available bed in selected ward type
  - Query: `SELECT bed_id FROM beds...`

|







### Error Handling:

- Transaction fails (connection lost) → ROLLBACK, show error toast, keep dialog open with data
- Concurrent booking (another user took the bed mid-transaction) → ROLLBACK, show “Bed just allocated to another patient”, refresh available beds
- Patient contact already exists → Ask “Patient already registered. Use existing record?”

### 3.3 Flow 3: Discharge Patient (Happy Path with Auto-Allocation)

```
[START: User on Dashboard]
|
|─ Path A: Click occupied (red) bed card
|
|─ Path B: Click "Discharge" in sidebar → Select from list
|
▼
```

```
1. Open Discharge Dialog
|
| - Show patient summary card
| - Pre-fill discharge date (today)
| - Default: "Make Available" selected
|
```

```
|
▼ User reviews and clicks "Confirm Discharge"
```

```
2. Show Confirmation Overlay
|
| - "Discharge [Patient Name] from
|   Bed [Bed Number]?"
| - "This action cannot be undone."
| - Buttons: "Yes, Discharge" | "Cancel"
|
```

```
|
▼ User confirms
```

```
3. Start DB Transaction
|
| - autoCommit = false
| - Isolation: REPEATABLE READ
|
```

```
|
▼
```

```
4. Update Admission record
|
| - status = 'Discharged'
| - actual_discharge = NOW()
|
```

```
|
▼
```

```
5. Update Bed Status
|
| - If "Make Available":
|   current_status = 'Available'
| - If "Maintenance":
|   current_status = 'Maintenance'
|   INSERT maintenance_log entry
|
```

```
|
▼ (Only if "Make Available")
```

6. Check Waitlist

```
- Query: SELECT * FROM waitlist  
WHERE requested_ward_type = ?  
AND status = 'Waiting'  
ORDER BY priority_score DESC,  
request_time ASC  
LIMIT 1
```

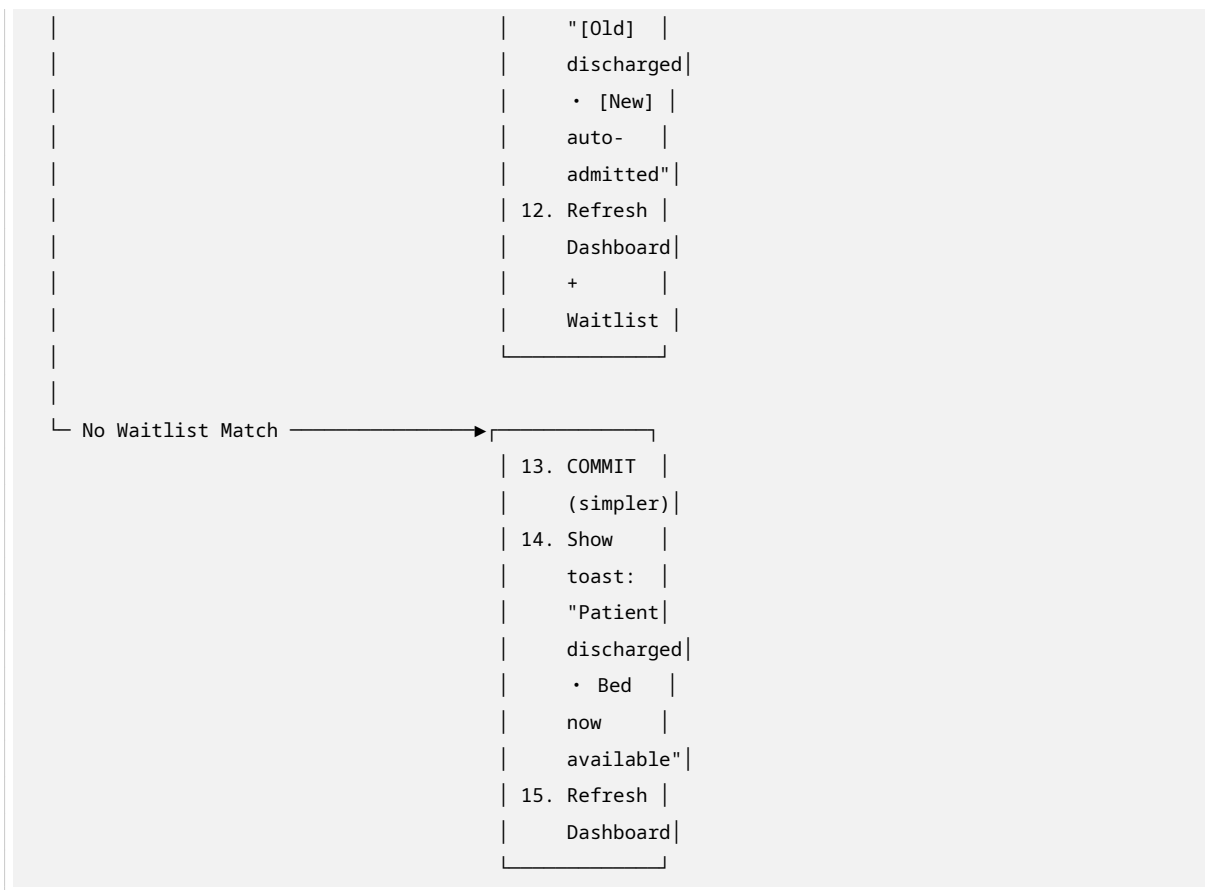
Waitlist Match Found

7. Auto-allocate:  
a. Update waitlist status= 'Allocated'  
b. Insert new admission for waitlist patient  
c. Update bed status= 'Occupied'

8. COMMIT transaction

9. Triggers fire for both discharge and new admission

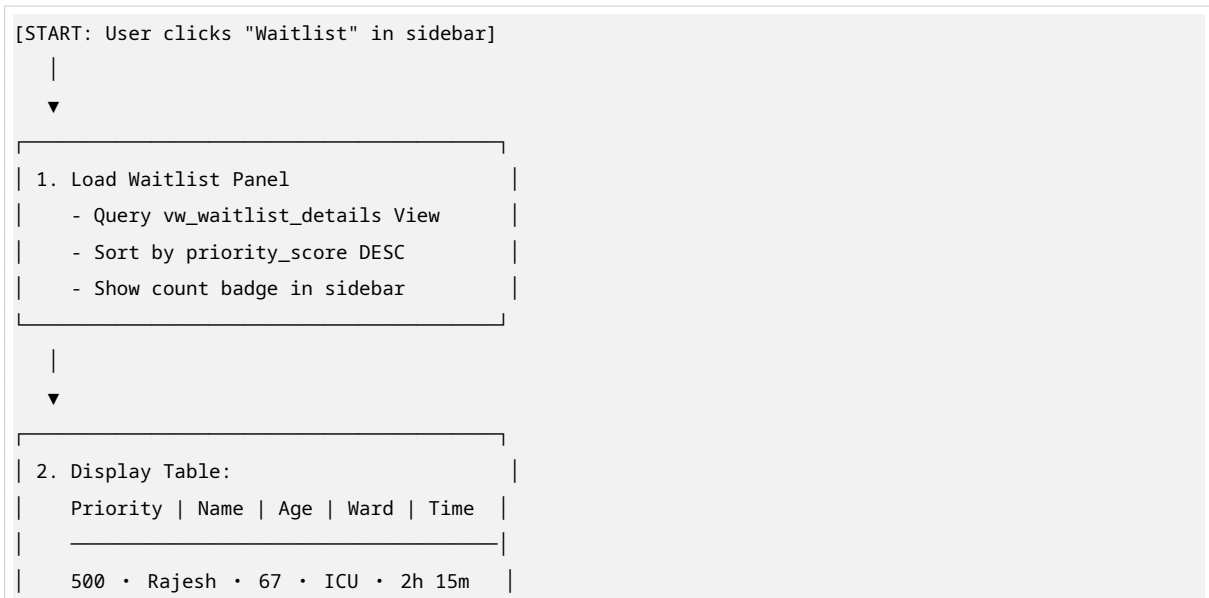
10. Close dialog  
11. Show success toast:



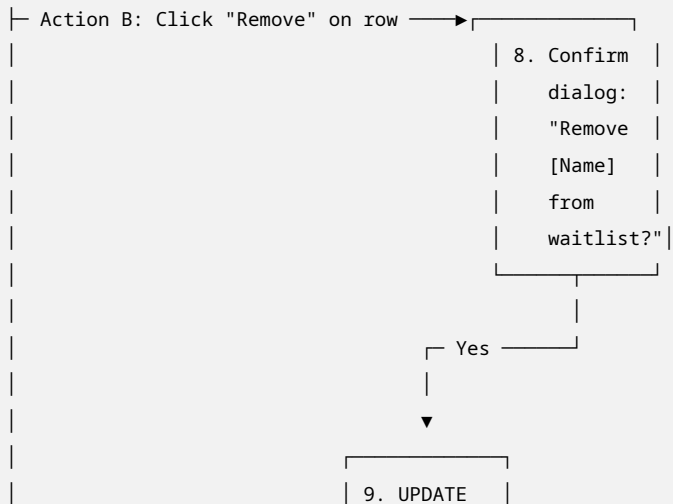
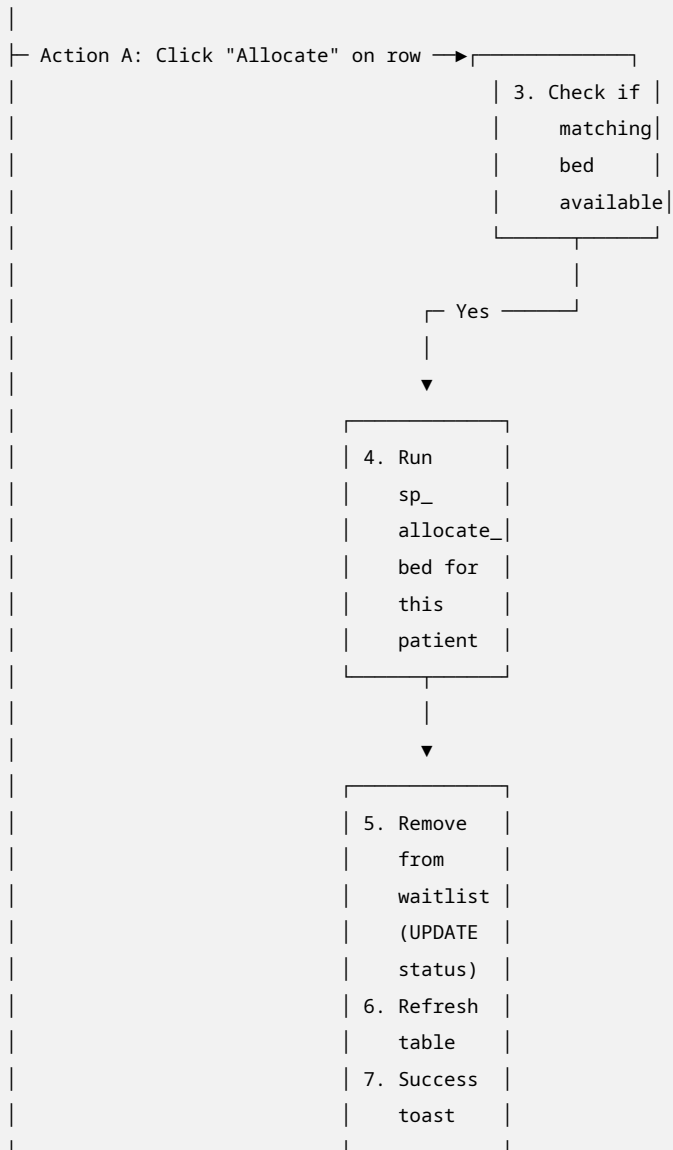
**Edge Cases:**

- User selects “Maintenance” → Skip waitlist check, bed unavailable until maintenance completed
- Maintenance completion → Admin manually marks “Available” → THEN check waitlist
- Concurrent discharge of same patient → Transaction isolation prevents conflict

**3.4 Flow 4: Waitlist Management**



|     |   |        |   |    |   |     |   |        |
|-----|---|--------|---|----|---|-----|---|--------|
| 400 | • | Sunita | • | 32 | • | ICU | • | 3h 40m |
| 300 | • | Rahul  | • | 45 | • | Gen | • | 5h 10m |
| ... |   |        |   |    |   |     |   |        |





### 3.5 Flow 5: Generate Report

```
[START: User clicks "Reports" in sidebar]
|
▼

1. Load Reports Center
- Show report category cards
- Each card: icon, title, desc

|
▼ User clicks "Occupancy Overview"

2. Load Report Detail View
- Back button (→ Reports Center)
- Title: "Occupancy Overview"
- Date range selector

|
▼ Default range: "Last 7 Days"

3. Query Database
- SELECT from vw_current_occupancy
- Or: Aggregate admissions table
- Group by ward_type, date range

|
▼

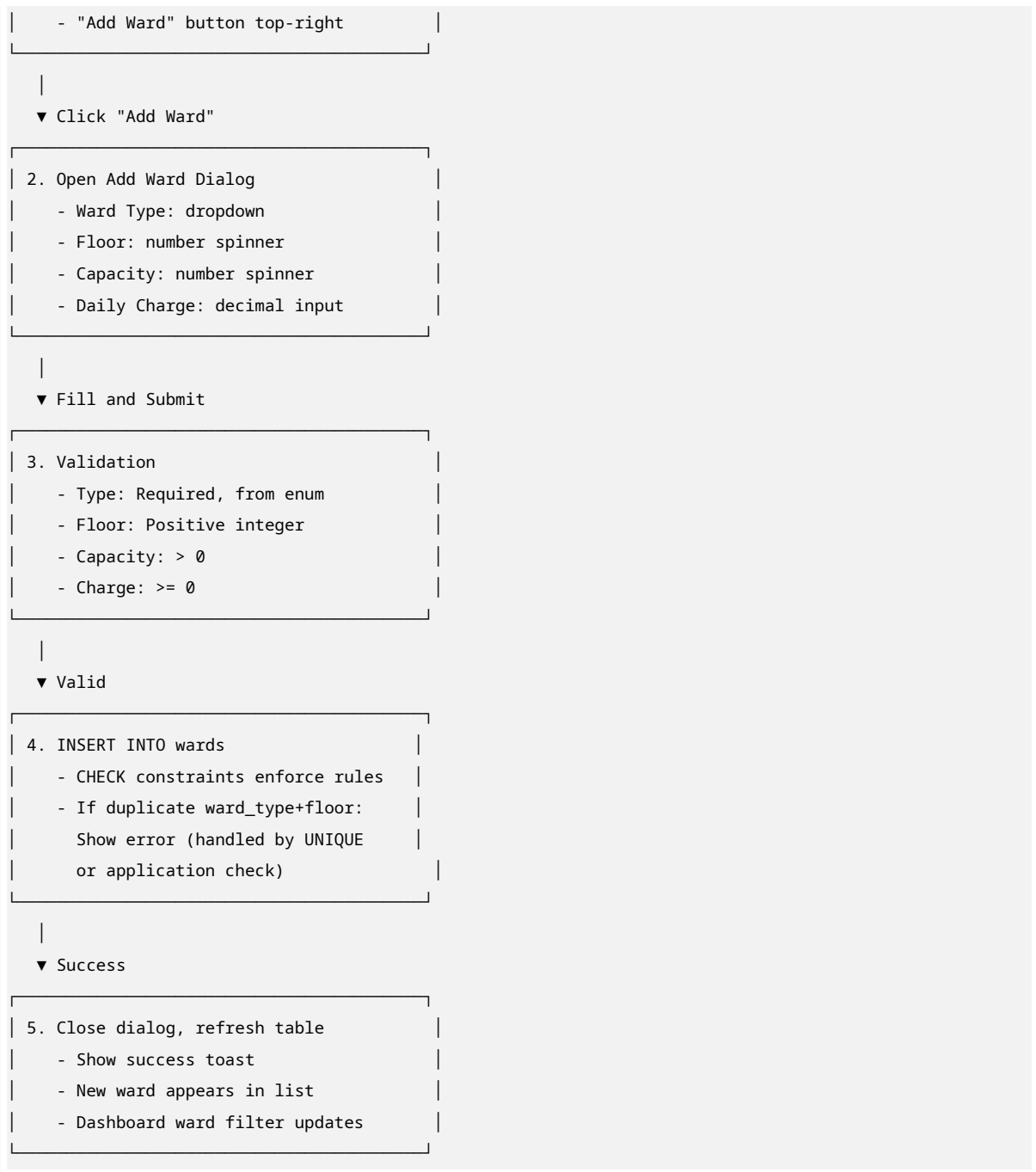
4. Render Results:
- Summary cards (3 metrics)
- Data table: Ward | Total | Occ | %
- Trend indicator (↑/↓ vs previous)
- Color-coded occupancy %
  (Green <70%, Yellow 70-90%,
   Red >90%)

|
└─ Action: Change date range → Re-run query, refresh table
|
└─ Action: Click "Export to CSV" → Generate CSV file, open save dialog
```

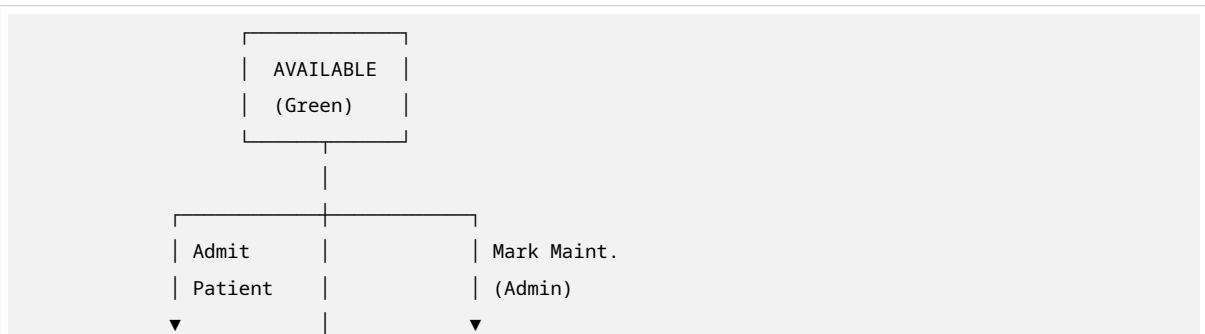
### 3.6 Flow 6: Admin — Add New Ward

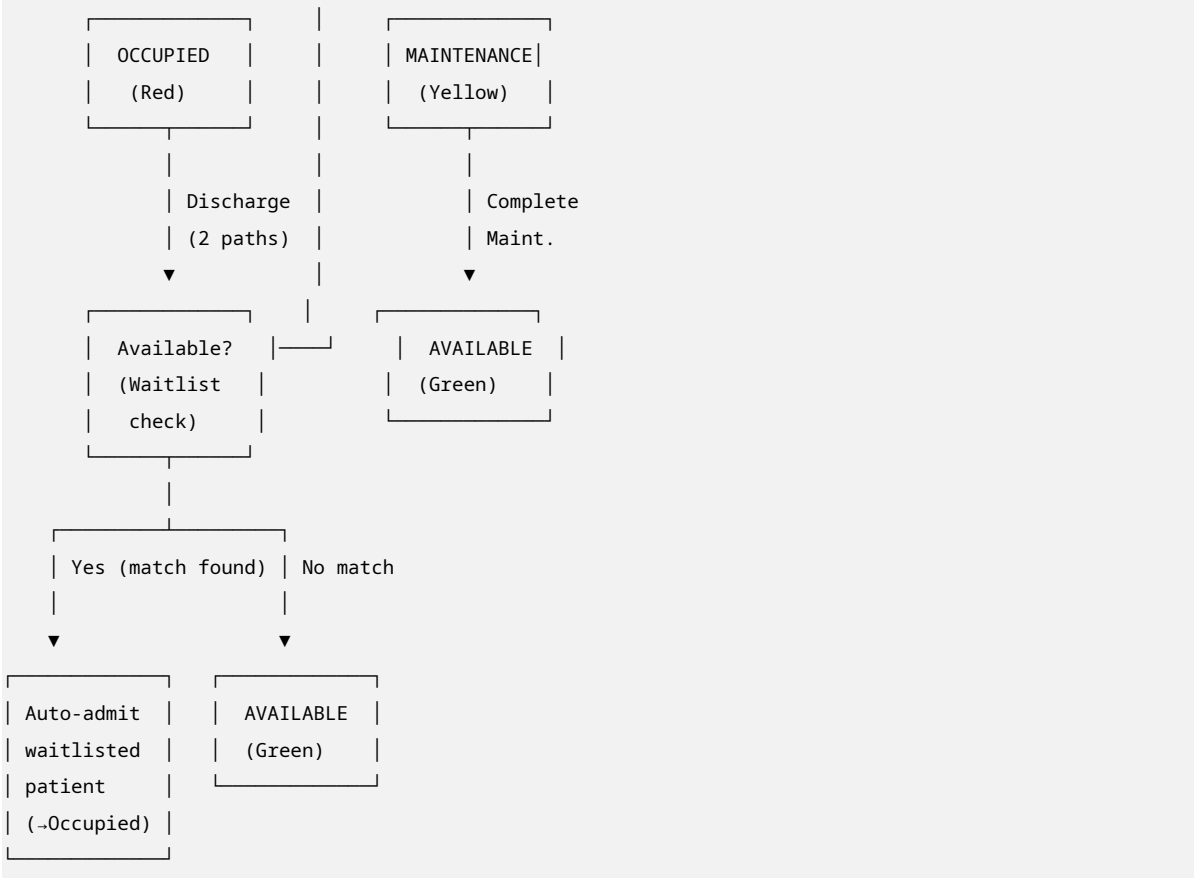
```
[START: Admin clicks "Admin Panel" → "Wards" tab]
|
▼

1. Display Wards Table
- Columns: ID, Type, Floor, Cap, etc
```



#### 4. State Machine: Bed Lifecycle



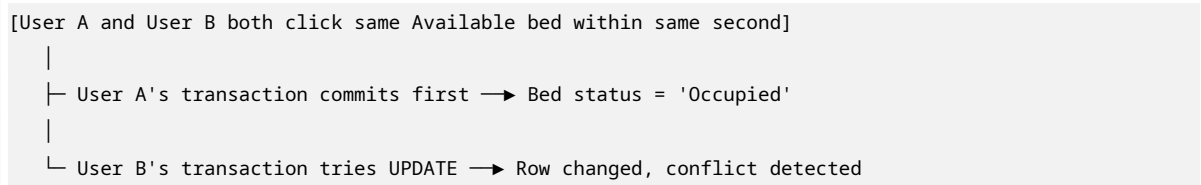


## 5. Error Flows

### 5.1 Database Connection Lost



### 5.2 Concurrent Bed Booking (Race Condition)



```

|
▼
| ROLLBACK |
| Show: "This |
| bed was just |
| allocated. |
| Refreshing |
| available |
| beds..." |
| Auto-refresh |
| bed grid |

```

5.3 Invalid Data Submission

```

[User submits form with errors]
|
▼
| 1. Client-side validation catches: |
| - Empty required fields |
| - Invalid formats (age = "abc") |
| - Logic errors (discharge < admit) |
| 2. Visual feedback: |
| - Red border on invalid fields |
| - Error text below each field |
| - First error field gets focus |
| 3. Form does NOT submit to database |

```

6. Notification & Feedback Matrix

| Action            | Success Feedback                  | Error Feedback                  | Duration    |
|-------------------|-----------------------------------|---------------------------------|-------------|
| Login             | None (direct to Dashboard)        | Red text below form + shake     | Until fixed |
| Admit Patient     | Toast: "Admitted to [Bed]"        | Dialog stays open, field errors | 4s          |
| Discharge Patient | Toast: "Discharged ·Bed [status]" | Dialog with retry               | 4s          |
| Add to Waitlist   | Toast: "Added to [Ward] waitlist" | Inline form error               | 4s          |
| Auto-Allocate     | Summary dialog with counts        | Toast: "No beds available"      | 5s          |
| Generate Report   | Table populates                   | Toast: "No data for range"      | 4s          |

Table 1 – continued

| Action             | Success Feedback                   | Error Feedback                    | Duration   |
|--------------------|------------------------------------|-----------------------------------|------------|
| Add Ward/Bed/Staff | Toast: “[Item] added successfully” | Dialog with field errors          | 4s         |
| DB Connection Lost | —                                  | Dialog: “Connection lost · Retry” | Persistent |

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*End of Application Flow Document*